Using Calculators to Find Averages

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Using Calculators to Find Averages

How do I use my graphic display calculator to find averages for raw data?

- You can use your **graphic display calculator** to help you to find the **quartiles** and **averages** for a set of **raw data**
  - For example, you may have been given a list of times that a group of 5 people took to run 100 m
- Enter the raw data into the first column of a **spreadsheet** on your graphic display calculator
  - It is a good idea to label the column (if possible) e.g. Time
  - Some calculator models will use a **statistics mode** instead, where you can enter the data in a **table**
- Select the statistics function and perform the statistics calculations for **one-variable statistics**
  - You need to make sure that the frequency is set to '1'
  - You may need to find the settings menu to change this
- Your graphic display calculator will display various **statistical measures** including:
  - Mean, $\bar{x}$
  - Median, MedianX or Med
  - Lower quartile, Q_1X or simply Q1
  - Upper quartile, Q_3X, or simply Q3

Examiner Tips and Tricks

- Make sure you know how to use the statistics function on your particular **graphic display calculator model**

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

calculator to find averages for data in a frequency table?

- You may be asked to find a particular average from a set of data that has been given to you in the form of a **frequency table**

Number of siblings	Frequency
0	4
1	8
2	6
3	2

- When entering data from a frequency table to a spreadsheet or table on your graphic display calculator, you need to use **two columns**
  - Enter the **data item** into the first column
    - Label the column, e.g. siblings
  - Enter the corresponding **frequency** into the second column
    - Label the column, e.g. freq
- Select the statistics function and perform the statistics calculations for **one-variable statistics**
  - You will need to specify which column contains the **data** and which contains the **frequency**
    - You may need to find the settings menu to change this
    - Some calculator models will refer to the columns as lists
- Your graphic display calculator will display various **statistical measures** including:
  - Mean, $\bar{x}$
  - Median, MedianX or Med
  - Lower quartile, Q_1X or simply Q1
  - Upper quartile, Q_3X, or simply Q3

Help me understand

Examiner Tips and Tricks

- If your calculator is able, labelling the columns in a spreadsheet is a good idea
  - This will make it easier to remember which is the **data** and which is the **frequency**

lower quartiles for the data set given below.

43 29 70 51 64 43

Enter the raw data as a list into a spreadsheet or table on your graphic display calculator
Label the list 'data'

data
43
29
70
51
64
43

Select the statistics calculations from the statistics function and choose one-variable statistics

Make sure the frequency is set to '1'

Read off the values for the median (MedianX or Med), the lower quartile (Q_1X or $Q1$) and the upper quartile (Q_3X or $Q3$)

Median = 47
Lower quartile = 43
Upper quartile = 64

How do I use my graphic display calculator to find averages for grouped data?

- You may be asked to find a particular average from a set of data that has been **grouped**

Length (cm)	Frequency
$0 < l \leq 5$	13
$5 < l \leq 10$	27
$10 < l \leq 15$	19
$15 < l \leq 20$	8

- You cannot enter a 'group', you must assume a value for each item in a particular group
  - **Assume** each data item in the group to be the **midpoint of the group**
- To find the midpoint of a group

$$\frac{5+10}{2} = 7.5$$

- It is useful to add the midpoint as an **extra column** in the table

Length (cm)	Frequency	Midpoint
$0 < l \leq 5$	13	2.5
$5 < l \leq 10$	27	7.5
$10 < l \leq 15$	19	12.5
$15 < l \leq 20$	8	17.5

- When entering data from a grouped frequency table to a spreadsheet on your graphic display calculator, you need to use **two columns**
  - Enter the **data item** into the first column
    - The data item will be values from the **midpoint column**
    - Label the column, e.g. length
  - Enter the corresponding **frequency** into the second column
    - Label the column, e.g. freq
- Select the statistics function and perform the statistics calculations for **one-variable statistics**
  - You will need to specify which column contains the **data** and which contains the **frequency**
    - You may need to find the settings menu to change this
    - Some calculator models will refer to the columns as lists
- Your graphic display calculator will display various **statistical measures** including:
  - Mean, $\bar{x}$
  - Median, MedianX or Med
  - Lower quartile, Q_1X or simply Q1
  - Upper quartile, Q_3X, or simply Q3

Examiner Tips and Tricks

- Whilst you can use your calculator to find the statistical measures, you should still **write down any supporting working**
  - E.g. Writing down the midpoints when dealing with grouped data can be worth a method mark
  - How many marks a question is worth can help indicate how much working is expected

Worked Example

A number of elephants drink at a particular waterhole.

The length of time that they spend at the waterhole one morning is recorded by a researcher and recorded in the following table:

Time, t (mins)	$0 \le t < 15$	$15 \le t < 30$	$30 \le t < 45$	$45 \le t < 60$
Frequency, f	5	8	6	2

Use your calculator to calculate an estimate for the mean time spent by an elephant at the waterhole.

Add another row to the table for the midpoints

Time, t (mins)	$0 \le t < 15$	$15 \le t < 30$	$30 \le t < 45$	$45 \le t < 60$
Frequency, f	5	8	6	2
Midpoint	$\frac{0+15}{2} = 7.5$	$\frac{15+30}{2} = 22.5$	$\frac{30+45}{2} = 37.5$	$\frac{45+60}{2} = 52.5$

Enter the data into a spreadsheet on your graphic display calculator
Add the 'Midpoint' values to the first column, label the list 'time'
Add the 'Frequency, f' values to the second column, label the list 'freq'

time	freq
7.5	5
22.5	8
37.5	6
52.5	2

Select the statistics calculations from the statistics function and choose one-variable statistics

Make sure the data is set as the 'time' column
Make sure the frequency is set to the 'freq' column

Read off the values for the mean ($\bar{x}$)

Mean = 26.07142...

Round to 3 significant figures

Mean = 26.1 minutes

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